

Proposal of a novel biomarker for neonatal seizure detection

First Author^{1,2*}, Second Author^{2,3†} and Third Author^{1,2†}

^{1*}Department, Organization, Street, City, 100190, State, Country.

²Department, Organization, Street, City, 10587, State, Country.

³Department, Organization, Street, City, 610101, State, Country.

*Corresponding author(s). E-mail(s): iauthor@gmail.com;
Contributing authors: iiauthor@gmail.com; iiiauthor@gmail.com;

[†]These authors contributed equally to this work.

Abstract

The abstract serves both as a general introduction to the topic and as a brief, non-technical summary of the main results and their implications. Authors are advised to check the author instructions for the journal they are submitting to for word limits and if structural elements like subheadings, citations, or equations are permitted.

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1 Introduction

Neonatal seizures represent the most frequent neurological emergency in the newborn period and are widely recognised as a hallmark of significant underlying brain dysfunction. With an estimated incidence of 1–5 per 1,000 live births in term neonates and substantially higher rates of 57–132 per 1,000 in preterm infants [?], seizures occur more commonly during the neonatal period than at any other stage of life. The predominant aetiologies include hypoxic–ischaemic encephalopathy, intracranial haemorrhage, ischaemic stroke, central nervous system infections, and inborn errors of metabolism [?]. Long-term follow-up studies have established that neonatal seizures are independently associated with adverse neurodevelopmental outcomes, predisposing affected infants to epilepsy, cerebral palsy, cognitive impairment, and behavioural

disorders in later childhood [?]. Because the prognosis depends critically on rapid identification and treatment, there is a compelling clinical need for accurate and timely seizure detection.

Electroencephalography (EEG) is the accepted gold standard for diagnosing neonatal seizures [?]. Scalp electrodes record the voltage differences produced by post-synaptic ionic currents, providing both the spatial and temporal resolution required to characterise seizure morphology, duration, and burden [?]. Continuous EEG monitoring in the neonatal intensive care unit has revealed that the majority of electrographic seizures—by some estimates 60–80%—are subclinical, exhibiting no outward clinical correlates and therefore escaping bedside recognition entirely [?]. Even when expert neurophysiologists review the recordings, inter-observer agreement remains only moderate; studies report significant discordance in seizure identification among clinicians with different levels of experience [?]. Manual interpretation is further constrained by its substantial time and labour requirements, rendering round-the-clock coverage impractical in many centres. These limitations collectively underscore the demand for automated, computer-aided seizure detection systems.

A diverse array of automated approaches has been investigated over the past two decades. Early methods relied on hand-crafted features drawn from the time, frequency, and time–frequency domains, which were subsequently classified by traditional machine learning algorithms such as support vector machines, random forests, and logistic regression [?]. Feature extraction strategies in this paradigm have included amplitude statistics, spectral band powers, wavelet coefficients, and autocorrelation measures [?]. Although these systems achieved reasonable sensitivity, they often suffered from high false-detection rates and limited generalisability across datasets and recording conditions.

More recently, deep learning architectures have been applied to neonatal seizure detection, offering end-to-end learning that can jointly optimise feature extraction and classification. Convolutional neural networks (CNNs) have demonstrated strong performance by automatically learning spatial and spectral representations from multichannel EEG [?]. Hybrid frameworks combining CNNs with recurrent layers, such as Long Short-Term Memory (LSTM) networks, have also been proposed; for example, the integration of Discrete Wavelet Transform decomposition with a 1D-CNN–LSTM pipeline has been shown to outperform traditional classifiers on several benchmark datasets [?]. In the few-shot learning setting, Empirical Mode Decomposition (EMD) has been used to decompose EEG signals into Intrinsic Mode Functions, enriching the time–frequency representation and enabling superior sensitivity even when training data are severely limited [?]. Despite these advances, the majority of deep learning methods operate as opaque end-to-end systems, offering limited interpretability regarding which physiological signal properties drive their decisions.

Parallel to the development of automated classifiers, a substantial body of work has investigated the spectral characteristics of the EEG as a source of physiologically meaningful biomarkers. Power spectral density (PSD) analysis quantifies how signal energy is distributed across frequency, and has been widely used to characterise brain-state changes during seizures. During neonatal seizures the PSD profile typically deviates from the characteristic $1/f^{-\alpha}$ power-law decay observed during normal

rest and sleep, with seizure epochs exhibiting elevated power in specific frequency bands, particularly in the theta (3–12 Hz) range [?]. Band-specific power measures—for instance, delta power and total spectral power—have been employed to predict encephalopathy severity and to identify neonates at risk for seizures within the first hour of EEG recording [?].

2 Methods

Our research has utilized parameters extracted from the curve fitting of log-transformed power spectral density as a feature set for classification which were then compared with a separate set of features obtained through Empirical Mode Decomposition.

2.1 Dataset

Our research used neonatal EEG data collected at Helsinki University Hospital. The cohort contained 79 infants, of whom 39 were confirmed to have seizures based on full-consensus annotation. Each EEG record spanned roughly an hour, with a median duration of 74 minutes (IQR 64–96 minutes). Seizure annotations were exported as text files containing onset times and durations at one-second resolution, incorporating information from all available EEG derivations. Because of this format, the exported annotations did not specify spatial characteristics or seizure type.

Only time segments in which all three expert annotators agreed on the presence of a seizure were retained as ground truth; any segments without unanimous agreement were removed from the dataset. The original referential montage was converted to a anterior-posterior bipolar configuration to match the montage used during the annotation process.

2.2 Log-transformed Power Spectral Density and Curve Fitting

Power spectral density provides a frequency-domain representation of signal energy, allowing broad trends in spectral shape to be described with compact quantitative measures.

A fourth-order Butterworth filter was applied to isolate the frequency components, allowing the signal to be separated into conventional brain-wave bands. This study focused specifically on the delta (0.7–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), and beta (13–30 Hz) ranges to provide cleaner representations of activity within these neurologically relevant frequencies.

After converting the spectrum to logarithmic scale for consistent scaling, linear regression was applied to model the overall spectral contour within the usable bandwidth. This approach produced a best-fit straight line capturing the dominant trend of the calibrated spectrum. From this line, three parameters were extracted for further analysis: the slope, the midband-fit value, and the intercept .

2.3 Empirical Mode Decomposition feature set

2.4 Evaluation

Principal Component Analysis was applied to reduce the two feature sets to a comparable dimensionality, allowing both pipelines to be evaluated under equivalent dimensional constraints. This dimensionality reduction step helped eliminate redundancy while preserving the dominant variance-bearing components within each feature space.

Following this, statistical comparisons were conducted using t-tests to assess the relevance of individual features, allowing an examination of which components differed meaningfully between seizure and non-seizure groups and providing insight into the discriminative value of the extracted vectors.

An artificial neural network was used as the classifier for both feature pipelines, where each system used the same feedforward architecture to ensure comparability. The network accepted a feature vector and passed it through a series of fully connected layers with 128, 64, and 32 units, respectively, with each layer followed by batch normalisation to stabilise activations, a ReLU nonlinearity to introduce non-linearity and a dropout rate of 0.3 to limit overfitting. The final layer consisted of a single output with a softmax function to produce class probabilities for seizure and non-seizure states. Training was performed using cross-entropy loss with class weights adjusted to compensate for the inherent imbalance between seizure and non-seizure samples, and the optimisation procedure used the Adam algorithm with a learning rate of 0.001 to update network parameters. Batches of 64 samples were used throughout training, and an early-stopping criterion monitored validation loss, stopping training if no improvement was observed within 10–25 epochs, with training capped at a maximum of 100 epochs.

The softmax output probabilities generated by the network were subsequently processed with a moving-average filter to reduce short-term fluctuations and produce a smoother probability trajectory. This temporal smoothing helped suppress brief disturbances or isolated spikes that did not reflect sustained activity. After filtering, an optimal decision threshold was determined to separate seizure from non-seizure predictions, allowing the system to convert the smoothed probability estimate into a final binary classification.

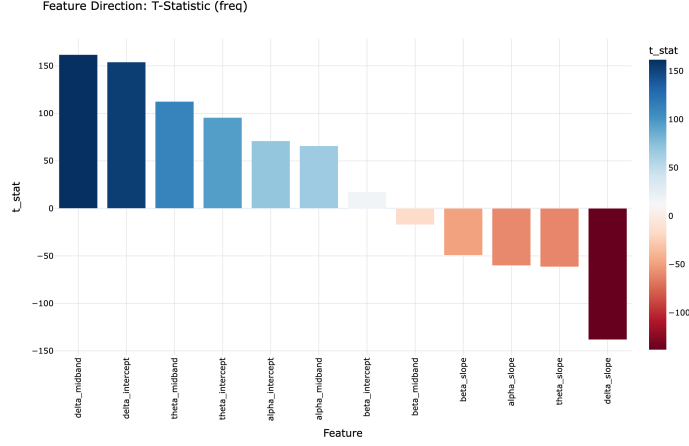
Neonatal EEG recordings exhibit severe class imbalance, as non-seizure epochs greatly exceed seizure epochs. To manage this, both classifiers employ an adaptive nearest-neighbour downsampling strategy designed to retain clinically relevant temporal structure. All seizure epochs are first preserved in full, ensuring that no positive examples are lost and then each non-seizure epoch is then assessed by measuring its temporal distance from the nearest seizure, providing a way to rank non-seizure samples based on their proximity to seizure activity.

Using this ranking, the method selects the closest non-seizure epochs according to a target ratio defaulted to the value 2.0. A minimum-ratio safeguard is applied so that non-seizure samples always exceed seizure samples numerically. Together, these steps reduce the degree of imbalance while retaining the temporal neighbourhood

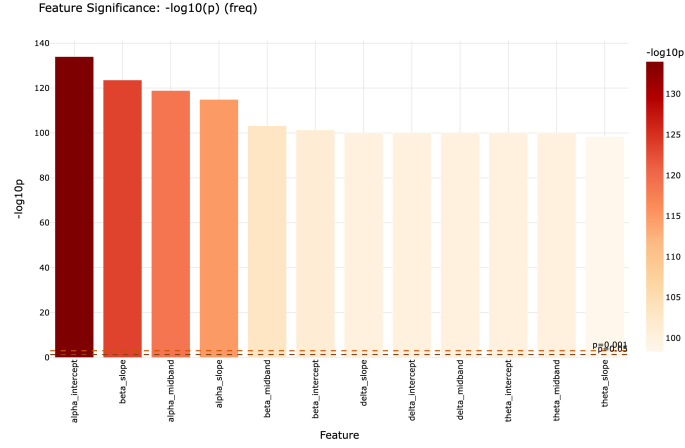
surrounding seizure events, preserving necessary context that may be important for classification.

3 Results

t-tests aggregate metrics thresholds and moving averages



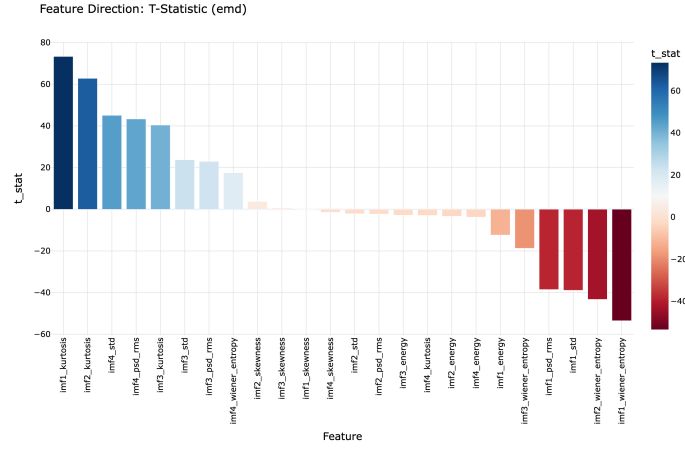
(a) Frequency Features : T-statistic



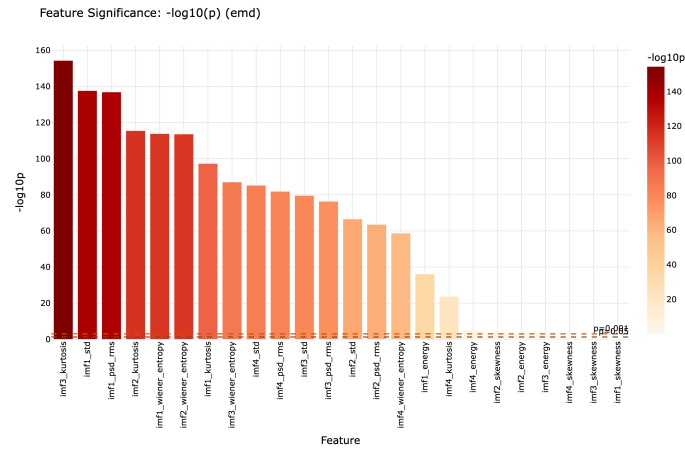
(b) Frequency Features : -log(p)

Fig. 1: Frequency-domain Results

Discussions should be brief and focused. In some disciplines use of Discussion or ‘Conclusion’ is interchangeable. It is not mandatory to use both. Some journals prefer



(a) EMD features : T-statistic



(b) EMD Features : $-\log(p)$

Fig. 2: EMD Results

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4 Conclusion

Conclusions may be used to restate your hypothesis or research question, restate your major findings, explain the relevance and the added value of your work, highlight any limitations of your study, describe future directions for research and recommendations.

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Authors reporting data from electrophoretic gels and blots should supply the full unprocessed scans for key as part of their Supplementary information. This may be requested by the editorial team/s if it is missing.

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Appendix A Section title of first appendix

An appendix contains supplementary information that is not an essential part of the text itself but which may be helpful in providing a more comprehensive understanding of the research problem or it is information that is too cumbersome to be included in the body of the paper.

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